

MIAE 215 · Rapid Review Sheet · Part 3

Expressions, Precedence, Modulo & Math Functions

1. Operator Precedence Hierarchy (Top to Bottom)

Precedence	Operators	Associativity	Key Rules & Notes
1 (Highest)	<code>() [] . -> ++ (post) -- (post)</code>	Left-to-Right	Postfix increments after expression evaluation
2	<code>+ - (unary) ++ (pre) -- (pre) ! sizeof</code>	Right-to-Left	Prefix increments before evaluation; <code>!0 == 1</code>
3	<code>* / %</code>	Left-to-Right	Multiplicative; <code>%</code> is integer only
4	<code>+ -</code>	Left-to-Right	Additive arithmetic
5	<code>< <= > >=</code>	Left-to-Right	Relational comparisons (yields <code>bool</code> 1 or 0)
6	<code>== !=</code>	Left-to-Right	Equality comparisons
7	<code>&&</code>	Left-to-Right	Logical AND (short-circuits on <code>false</code>)
8	<code> </code>	Left-to-Right	Logical OR (short-circuits on <code>true</code>)
9 (Lowest)	<code>= += -= *= /= %=</code>	Right-to-Left	Chained: <code>x = y = z = 10;</code> assigns right-to-left

2. Modulo (%) & Discrete Engineering Patterns

- **Digit Peeling (Reversal):** `digit = n % 10; n /= 10;` (peels digits from right to left).
 - Example: For 237, successive remainders are 7, 3, 2.
- **Cyclic Buffer:** `idx = (idx + 1) % N;` wraps index safely within $[0, N - 1]$.
- **Parity & Sign Flip:** `(k % 2 == 0) ? 1 : -1;` generates alternating series $(-1)^k$.

3. Division & Multi-Assignment Traps

- **Integer Truncation:** `1 / 3` is strictly `0`! Thus, `1 / 3 * 10.0 == 0.0`.
 - **Fix:** Force floating-point literal: `1.0 / 3.0 * 10.0 == 3.333333333333333`.
- **Right-Associative Assignment:** `int x, y, z; x = y = z = 5;` evaluates `z=5`, then `y=5`, then `x=5`.
- **Postfix in Expressions:** `q = 10; int r = q++;` $\implies r = 10, q = 11$.

4. <cmath> Functions & IEEE 754 Precision Quirks

- **Angle Units:** All trigonometric arguments must be in **radians** ($\text{rad} = \text{deg} \times \frac{\pi}{180}$).
- **Trig Functions:** `sin(x)`, `cos(x)`, `tan(x)`, `asin(x)`, `acos(x)`, `atan2(y, x)`.
- **Powers & Roots:** `pow(base, exp)`, `sqrt(x)`, `exp(x)`, `log(x)` (natural $\ln$), `log10(x)`.
- **Floating-Point Noise:** `cos( $\pi/2$ )` evaluates to $\approx 6.12323 \times 10^{-17}$ (NOT 0.0).
- **Catastrophic Cancellation:** `1.234567890123456 - 1.234567890123455` = 1.0×10^{-15} (retains 1 digit).
- **Machine Epsilon:** For 64-bit `double`, $\epsilon \approx 2.22 \times 10^{-16}$. Thus, `1.0 - (1.0 - 10-20)` = 0.0.